Mathematical Thinking for Number Problems (Pen & Paper First → Python Next)

🎯 Objective

The purpose of this task is **NOT** to directly write Python programs.

The goal is to train your brain to **think mathematically first**, exactly like solving problems with **pen and paper**, and then convert that mathematical logic into Python code.

> **Remember:** Programming is nothing but converting your mathematical or logical thinking into code.

If you don't understand the mathematics behind a problem, you will only memorize Python syntax instead of learning problem-solving.

📌 Important Rule

For **every question**, follow the steps **in the exact order** given below.

Step 1 – Understand the Mathematical Logic (Pen & Paper)

Before writing any code,

- Write the complete mathematical process.
- Solve the problem manually using pen and paper.
- Explain how each step works mathematically.
- Show how the value changes after every mathematical operation.

Do **NOT** write Python code in this step.

Step 2 – Dry Run

Take at least one example.

Write every intermediate step.

Example:

Number = 153

Last Digit = ?

Remaining Number = ?

Current Answer = ?

Next Iteration = ?

Final Answer = ?

Every calculation should be shown clearly.

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Step 3 – Identify the Pattern

After completing the mathematical solution,

answer these questions:

- What operation is repeated?
- Which mathematical operators are used?
- How do we move from one digit to another?
- What is the stopping condition?
- Which variables are required?

Step 4 – Write the Python Program

Only after completing the first three steps,

write the Python program.

🚫 Restrictions

You are **NOT ALLOWED** to use any string-based solution.

Do **NOT** use

str()
len(str())
list()
tuple()
[::-1]
reversed()
join()

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Use only mathematical operations such as

+
-
*
/
//
%
**

📖 Example

Suppose the question is

Reverse a Number

Number

153

Do **NOT** immediately write Python code.

Instead,

First

Write how you would solve it manually on paper.

153

↓

Take last digit

↓

Store it

↓

Remove last digit

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↓

Repeat

↓

Final Answer

Then identify

- Which operator extracts the last digit?
- Which operator removes the last digit?
- How is the reversed number formed?

Only then write the Python code.

📝 Questions

Part A — Reverse Number

Question 1

Reverse a given number.

Example

153

Output

351

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Question 2

Reverse a number and check whether it is a palindrome.

Example

121

Output

Palindrome

Question 3

Reverse only the last **N** digits of a number.

Example

123456

N = 3

Output

123654

Part B — Count Digits

Question 4

Find the total number of digits in a number.

(Without using `len()` or converting the number into a string.)

Question 5

Count the total even digits in a number.

Example

824731

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Output

3

Question 6

Count the total odd digits in a number.

Example

824731

Output

3

Part C — Digit Extraction

Question 7

Find the first digit of a number.

Question 8

Find the last digit of a number.

Question 9

Find the sum of all digits.

Question 10

Find the product of all digits.

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📋 Submission Format (Mandatory)

For **every question**, your submission should contain the following sections:

✅ Section 1 — Mathematical Logic (Pen & Paper)

- Solve the problem completely using mathematics.
- Show every intermediate calculation.
- Explain how the answer is obtained manually.
- Do **NOT** write Python code.

✅ Section 2 — Dry Run

Take an example input.

Show every iteration and every mathematical calculation until the final answer.

✅ Section 3 — Logic Building

Write in your own words:

- What mathematical pattern did you observe?
- Which operations are repeated?
- Which variables are required?
- What is the stopping condition?
- How can this mathematical logic be converted into an algorithm?

✅ Section 4 — Python Program

Finally, convert your mathematical logic into a Python program.